Chapter 10: One- and Two-Sample Tests of Hypotheses (Part II)

ENGR 0021

Learning Objectives

- Single-sample tests concerning a single mean
- Two-sample tests on two means
- Choice of sample size for testing means
- One- and two-sample tests concerning variances

Single Sample: Tests Concerning a Single Mean

- Assumptions:
  - $X_1, X_2, \ldots, X_n$ is a random sample from a population with mean μ and variance $\sigma^2 > 0$
- Testing procedure for a single mean, variance known:
  - Normal population
  - Any population, large-sample test by the CLT
- Testing procedure for a single mean, variance unknown:
  - Normal population, small-sample test using the t distribution
  - Any population, large-sample test by the CLT

Test Procedure for a Single Mean (Variance Known)

$$H_0 : \mu = \mu_0$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu \neq \mu_0 \iff |z| \geq z_{\alpha/2}$$
$$H_1 : \mu > \mu_0 \iff z \geq z_{\alpha}$$
$$H_1 : \mu < \mu_0 \iff z \leq -z_{\alpha}$$

Relationship to Confidence Interval Estimation

- Confidence interval estimation computes bounds within which it is reasonable for the parameter to lie.
- Consider the test at significance level α:

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0$$

- This test is equivalent to computing a $100(1-\alpha)\%$ confidence interval for μ and rejecting H_0 if μ_0 lies outside the interval.

Tests on a Single Sample (Variance Unknown)

If $X_1, \ldots, X_n$ is a random sample from a normal distribution, then

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

Test Procedure for a Single Mean (Variance Unknown)

$$H_0 : \mu = \mu_0$$

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu \neq \mu_0 \iff |t| \geq t_{\alpha/2,n-1}$$
$$H_1 : \mu > \mu_0 \iff t \geq t_{\alpha,n-1}$$
$$H_1 : \mu < \mu_0 \iff t \leq -t_{\alpha,n-1}$$

Example: Light Bulb Lifetime

Light bulbs are advertised as having an average lifetime of 750 hours. A customer will purchase them unless it can be shown that the true mean lifetime is smaller.

From a random sample of 10 bulbs:

$$\bar{x} = 738.44, \qquad s = 38.2$$

Assume a normal population. Test at significance level $\alpha = 0.05$:

$$H_0 : \mu = 750 \qquad H_1 : \mu < 750$$

Example: Light Bulb Lifetime

Two Samples: Tests on Two Means

- Independent samples, known variances
  - Normal populations
  - Large samples
- Normal populations, unknown variances
  - Variances unknown and equal
  - Variances unknown and unequal
- Paired observations

Two Samples: Test on Two Means

If $\bar{X}_1$ and $\bar{X}_2$ are means of two independent random samples of sizes n_1 and n_2 from populations with means μ_1 and μ_2 and known variances σ_1^2 and σ_2^2, then

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

has a standard normal distribution.

- Exact for normal populations
- Approximate for large samples by the CLT

Test Procedures for Normal Populations with Known Variances

$$H_0 : \mu_1 - \mu_2 = d_0$$

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu_1 - \mu_2 \neq d_0 \iff |z| \geq z_{\alpha/2}$$
$$H_1 : \mu_1 - \mu_2 > d_0 \iff z \geq z_\alpha$$
$$H_1 : \mu_1 - \mu_2 < d_0 \iff z \leq -z_\alpha$$

Test Procedures for Large Samples

$$H_0 : \mu_1 - \mu_2 = d_0$$

If variances are known,

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

If variances are unknown,

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu_1 - \mu_2 \neq d_0 \iff |z| \geq z_{\alpha/2}$$

$$H_1 : \mu_1 - \mu_2 > d_0 \iff z \geq z_\alpha$$

$$H_1 : \mu_1 - \mu_2 < d_0 \iff z \leq -z_\alpha$$

Two-Sample t-Test: Unknown but Equal Variances

$$H_0 : \mu_1 - \mu_2 = d_0$$

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_\nu, \qquad \nu = n_1 + n_2 - 2$$

where

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu_1 - \mu_2 \neq d_0 \iff |t| \geq t_{\alpha/2,\nu}$$

$$H_1 : \mu_1 - \mu_2 > d_0 \iff t \geq t_{\alpha,\nu}$$

$$H_1 : \mu_1 - \mu_2 < d_0 \iff t \leq -t_{\alpha,\nu}$$

Two-Sample t-Test: Unknown and Unequal Variances (Welch's t-test)

$$H_0 : \mu_1 - \mu_2 = d_0$$

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \sim t_\nu$$

with approximate degrees of freedom

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1 - 1) + (S_2^2/n_2)^2/(n_2 - 1)}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu_1 - \mu_2 \neq d_0 \iff |t| \geq t_{\alpha/2,\nu}$$

$$H_1 : \mu_1 - \mu_2 > d_0 \iff t \geq t_{\alpha,\nu}$$

$$H_1 : \mu_1 - \mu_2 < d_0 \iff t < -t_{\alpha,\nu}$$

Example: GPA and Car Ownership

Two random samples of 100 male students each were selected.

Non-owners:

$$n_1 = 100, \quad \bar{x}_1 = 2.70, \quad \sigma_1^2 = 0.36$$

Car owners:

$$n_2 = 100, \quad \bar{x}_2 = 2.54, \quad \sigma_2^2 = 0.40$$

Test at $\alpha = 0.05$ whether the mean GPA differs:

$$H_0 : \mu_1 - \mu_2 = 0, \qquad H_1 : \mu_1 - \mu_2 \neq 0$$

Example: GPA and Car Ownership

Example: Ultrasonic Humidifiers (Pooled t CI)

Brand 1 outputs (oz): 14.0, 14.3, 12.2, 15.1
Brand 2 outputs (oz): 12.1, 13.6, 11.9, 11.2
Are the brand outputs different? Use $\alpha = .05$.

Example: Fabric Porosity

A study compares air permeability for two types of plain-weave fabric.

Fabric Type	Sample Size	Sample Mean	Sample Std Dev
Cotton	10	51.71	0.79
Triacetate	10	136.14	3.59

Assuming normality and unequal variances, test whether the true average porosity of triacetate exceeds that of cotton using $\alpha = .05$.

The Paired t-Test

Null hypothesis:

$$H_0 : \mu_D = d_0$$

Test statistic:

$$T = \frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$$

where:

- $\bar{D}$ is the sample mean of the differences
- S_D is the sample standard deviation of the differences
- n is the number of pairs

The differences are assumed to be normally distributed.

Paired t-Test Rejection Regions

$$H_0 : \mu_D = d_0$$

$$T = \frac{\bar{D} - d_0}{S_D/\sqrt{n}} \sim t_{n-1}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \mu_D \neq d_0 \iff |t| \geq t_{\alpha/2,n-1}$$
$$H_1 : \mu_D > d_0 \iff t \geq t_{\alpha,n-1}$$
$$H_1 : \mu_D < d_0 \iff t \leq -t_{\alpha,n-1}$$

Example: Fabric Breaking Load

Breaking load (kg/25 mm width) was measured for fabrics in both unabraded and abraded conditions.

Fabric	1	2	3	4	5	6	7	8
Unabraded	36.4	55.0	51.5	38.7	43.2	48.8	25.6	49.8
Abraded	28.5	20.0	46.0	34.5	36.5	52.5	26.5	46.5
Difference	7.9	35.0	5.5	5.2	6.7	-3.7	-0.9	3.3

Does the data suggest that unabraded fabric has a greater breaking load than abraded fabric? Use $\alpha = 0.01$.

Pros and Cons of Pairing

- The choice between an independent two-sample t test and a paired t test is not always straightforward.
- One tradeoff is the number of degrees of freedom.
- In a paired t test, we typically lose degrees of freedom.
- Higher degrees of freedom are usually preferred because they improve precision.

When is Pairing Helpful?

- If experimental units are highly heterogeneous, a two-sample test may have difficulty detecting small but significant differences.
- Pairing is preferred when:
  - heterogeneity is large, and
  - correlation within pairs is large
- If units are relatively homogeneous and correlation within pairs is small, independent samples are preferred.
- Pairing is a special case of **blocking**.

Summary: Choosing the Right Test

- Step 1: One sample or two samples?
- Step 2: Are the samples paired?
  - Yes $\rightarrow$ Paired t test
  - No $\rightarrow$ continue
- Step 3: Is the population variance known?
  - Yes $\rightarrow$ z test
  - No $\rightarrow$ t test
- Step 4 (two samples only): Are variances equal?
  - Equal $\rightarrow$ pooled t test
  - Unequal $\rightarrow$ Welch t test

Choice of Sample Size for Testing Means: One Sample

For a fixed significance level α and fixed power $1-\beta$, for the alternative

$$\mu = \mu_0 + \delta$$

the approximate required sample size is:

$$H_1 : \mu \neq \mu_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2}$$

$$H_1 : \mu > \mu_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}$$

$$H_1 : \mu < \mu_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}$$

Choice of Sample Size for Testing Means: Two Samples

For a fixed significance level α and fixed power $1-\beta$, for the alternative

$$\mu_1 - \mu_2 = d_0 + \delta$$

and equal sample sizes $n = n_1 = n_2$, the approximate required sample size is:

$$H_1 : \mu_1 - \mu_2 \neq d_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_{\alpha/2} + z_\beta)^2(\sigma_1^2 + \sigma_2^2)}{\delta^2}$$

$$H_1 : \mu_1 - \mu_2 > d_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_{\alpha} + z_\beta)^2(\sigma_1^2 + \sigma_2^2)}{\delta^2}$$

$$H_1 : \mu_1 - \mu_2 < d_0 \quad \Longleftrightarrow \quad n \approx \frac{(z_{\alpha} + z_\beta)^2(\sigma_1^2 + \sigma_2^2)}{\delta^2}$$

One-Sample Test Concerning Variance

Review:

- We estimate σ^2 using S^2.
- The statistic
$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$
has a chi-squared distribution with $n-1$ degrees of freedom.

Hypothesis Test for a Population Variance

$$H_0 : \sigma^2 = \sigma_0^2$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

Alternative hypothesis $\iff$ rejection region:

$$H_1 : \sigma^2 > \sigma_0^2 \iff \chi^2 > \chi^2_{\alpha,n-1}$$
$$H_1 : \sigma^2 < \sigma_0^2 \iff \chi^2 < \chi^2_{1-\alpha,n-1}$$
$$H_1 : \sigma^2 \neq \sigma_0^2 \iff \chi^2 < \chi^2_{1-\alpha/2,n-1} \text{ or } \chi^2 > \chi^2_{\alpha/2,n-1}$$

Two-Sample Test Concerning Variances

Review:

- We estimate the ratio σ_1^2/σ_2^2 using S_1^2/S_2^2.
- The statistic
$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$
has an F distribution with $n_1 - 1$ and $n_2 - 1$ degrees of freedom.

F Test for Equality of Variances

$$H_0 : \sigma_1^2 = \sigma_2^2 \qquad \Longleftrightarrow \qquad \frac{\sigma_1^2}{\sigma_2^2} = 1$$

$$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1,n_2-1}$$

Alternative hypothesis $\Longleftrightarrow$ rejection region:

$$H_1 : \sigma_1^2 > \sigma_2^2 \Longleftrightarrow f > F_\alpha(n_1 - 1, n_2 - 1)$$
$$H_1 : \sigma_1^2 < \sigma_2^2 \Longleftrightarrow f < F_{1-\alpha}(n_1 - 1, n_2 - 1)$$
$$H_1 : \sigma_1^2 \neq \sigma_2^2 \Longleftrightarrow f < F_{1-\alpha/2}(n_1 - 1, n_2 - 1) \text{ or } f > F_{\alpha/2}(n_1 - 1, n_2 - 1)$$

Example: Variability in Rat Weight Gain

To investigate the effect of an insecticide, weight gains were measured for:

- 23 female control rats: sample standard deviation $= 32$ g
- 20 female low-dose rats: sample standard deviation $= 54$ g

Does the data suggest that variability in low-dose weight gains is greater than in control weight gains?

Assume normality and test at significance level $\alpha = 0.05$:

$$H_0 : \sigma^2_{\text{low}} = \sigma^2_{\text{control}} \qquad H_1 : \sigma^2_{\text{low}} > \sigma^2_{\text{control}}$$

Summary: Choosing the Right Test

Decision

1. Means or variances?
2. One or two samples?
3. Paired?
4. Known σ?
5. Equal variances?

Tests

- Means: z, t, Welch, paired t
- Variance: χ^2, F

Most Common

- One sample $\rightarrow$ t
- Two samples $\rightarrow$ Welch t

Key

$$\text{test statistic} = \frac{\text{signal}}{\text{noise}}$$

Summary Table (Table 10.3)

H_0	Value of Test Statistic	H_1	Critical Region
$\mu = \mu_0$	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$; σ known	$\mu < \mu_0$ $\mu > \mu_0$ $\mu \neq \mu_0$	$z < -z_\alpha$ $z > z_\alpha$ $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
$\mu = \mu_0$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$; $v = n - 1$, σ unknown	$\mu < \mu_0$ $\mu > \mu_0$ $\mu \neq \mu_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2}$ or $t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$z = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$; σ_1 and σ_2 known	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$z < -z_\alpha$ $z > z_\alpha$ $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p\sqrt{1/n_1 + 1/n_2}}$; $v = n_1 + n_2 - 2$, $\sigma_1 = \sigma_2$ but unknown, $s_p^2 = \dfrac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2}$ or $t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t' = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$; $v = \dfrac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$, $\sigma_1 \neq \sigma_2$ and unknown	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$t' < -t_\alpha$ $t' > t_\alpha$ $t' < -t_{\alpha/2}$ or $t' > t_{\alpha/2}$
$\mu_D = d_0$ paired observations	$t = \dfrac{\bar{d} - d_0}{s_d/\sqrt{n}}$; $v = n - 1$	$\mu_D < d_0$ $\mu_D > d_0$ $\mu_D \neq d_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2}$ or $t > t_{\alpha/2}$

Interpreting the p-value

We run a two-sample t-test for equal means with $\alpha = 0.05$ and obtain a p-value of 0.04.

Question: What are the odds that the two samples are drawn from distributions with the same mean?

(a) 19/1

(b) 1/19

(c) 1/20

(d) 1/24

(e) unknown

Children IQ Test

An investigator wants to show that first-born children score higher on IQ tests than second-borns. He takes a simple random sample of 400 two-child families in a school district, with both children enrolled in elementary school.
He administers the WISC vocabulary test (scores corrected for age differences) and obtains the following results:

- The 400 first-borns have a sample mean of 29 and a sample standard deviation of 10.
- The 400 second-borns have a sample mean of 28 and a sample standard deviation of 10.

He performs a two-sample z-test:

$$\text{SE for first-born mean} \approx 0.5$$
$$\text{SE for second-born mean} \approx 0.5$$
$$\text{SE for difference} = \sqrt{0.5^2 + 0.5^2} \approx 0.7$$
$$z = \frac{1}{0.7} \approx 1.4, \quad p \approx 0.08$$

Comment briefly on the use of statistical tests in this situation.